HYEONJAE JEON, Ph.D.

Curriculum Vitae

November 2, 2026

■ Research Profile

Researcher in autonomous driving perception, mobile robotic systems, and deep learning-based visual robustness. My doctoral research investigated how degraded visual observations affect perception and behavior, and developed analysis, data generation, and restoration methods for robust autonomous systems. I am currently a postdoctoral researcher at DGIST (InnoCORE Emergent Intelligence Physical AI Research Center), where my research focuses on Vision-Language-Action (VLA) models and their real-world deployment on humanoid robots—in particular, perturbation-robust VLA post-training through movement-primitive-based recovery data generation that requires no simulator rollouts—along with imitation learning, embodied AI, and world-model-based robot learning for humanoid and dual-arm robots.

■ Research Interests

- Vision-Language-Action (VLA) models, imitation learning, and embodied AI for humanoid and dual-arm robot systems, including post-training and real-world deployment of VLA policies on humanoid robots.
- Perturbation-robust VLA policies: movement-primitive (DMP)-based recovery data generation without simulator rollouts, analysis of failure modes under out-of-distribution robot states, and inference-time instruction grounding.
- Robust perception and action under imperfect visual observations, including occlusion, adverse weather, sensor blockage, blur, and low-light conditions.
- Task-oriented data generation, diffusion-based image restoration, and perception-aware representation learning for real-robot generalization.
- Autonomous driving and mobile robot perception/control, including lane detection, path tracking, visual servoing, and ROS-based robotic system implementation.
- World-model-based robot learning with task-relevant visual state prediction, uncertainty modeling, and practical deployment-oriented evaluation.

■ Personal

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■ Academic Degrees

Ph. D. Department of Robotics and Mechatronics Engineering, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Korea, August 2026. Dissertation: Study on Degradation of Autonomous Driving Performance in Adverse Weather Conditions and Deep Learning-based Improvement.

B.S. School of Undergraduate Studies, College of Transdisciplinary Studies, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Korea, February 2021.

■ Professional Accomplishments: Research Projects

(R6) Autonomous Developmental Whole-Body Control Framework Inspired by Human Behavioral Learning Mechanisms, 2026.3 – 2031.2, (Sponsor: National Research Foundation of Korea, NRF, Korea)

(R5) Establishment of experimental environment and dataset for driving control of 4-wheel independent steering vehicle, 2024.9 – 2025.4, (Sponsor: Electronics and Telecommunications Research Institute, ETRI, Daegu, Korea)

(R4) Analysis of safety range of autonomous driving system operation according to driving environment, 2024.7 – 2024.10, (Sponsor: Korea Intelligent Automotive Parts Promotion Institute, KIAPI)

(R3) Development of Lane Perception and Path Tracking Control Algorithm for Self-Driving Car, 2023.5 – 2023.12, (Sponsor: Electronics and Telecommunications Research Institute, ETRI, Daegu, Korea)

(R2) Development of blind spot path planning/tracking algorithm of autonomous mobile robots using multi-sensor integration for efficient manufacturing automation, 2021.7 – 2023.12, (Sponsor: Ministry of Science and ICT, Korea)

(R1) Development of Evaluation Technology for V2E Cognitive Judgment Safety and Accident Response. 2021.4 – 2026.12, (Sponsor: Korea Agency for Infrastructure Technology Advancement, KAIA, Korea)

■ Publications: Journal Papers

International Journal

(J5) **Hyeonjae Jeon**, Yongseob Lim, “Severity-Aware Cross-Attention Diffusion for Multi-Degradation Image Restoration”, *Pattern Recognition*, 2026. Manuscript under review.

(J4) **Hyeonjae Jeon**, Junghyun Seo, Taesoo Kim, Sungho Son, Jungki Lee, Gyeungho Choi, Yongseob Lim, “RainSD: Rain Style Diversification Module for Image Synthesis Enhancement using Feature-Level Style Distribution”, *Robotics and Autonomous Systems, Vol. 186*, 2025.

(J3) Junghyun Seo, Sungjun Wang, **Hyeonjae Jeon**, Yongsik Jin, Soon Kwon, Jeseok Kim, Yongseob Lim, "LuminanceGAN: Controlling the brightness of generated images for data augmentation in various night conditions", *Pattern Recognition Letters, Vol. 186*, 2024.

(J2) Sungjun Wang, Junghyun Seo, **Hyunjae Jeon**, Sanghyun Park, Yongseob Lim, "Horizontal Attention Based Generation Module for Unsupervised Domain Adaptive Stereo Matching", *IEEE Robotics and Automation Letters (RA-L),* 2023, DOI: 10.1109/LRA.2023.3313009.

(J1) **Hyeonjae Jeon**, YaeOhn Kim, Minyoung Choi, Donggeon Park, Sungho Son, Jungki Lee, Gyeungho Choi, Yongseob Lim, "CARLA Simulator-based Evaluation Framework Development of Lane Detection Accuracy Performance under Sensor Blockage caused by Heavy Rain for Autonomous Vehicle", *IEEE Robotics and Automation Letters (RA-L),* 2022, DOI: 10.1109/LRA.2022.3192632.

■ Publications: Conference Proceedings and Presentations

International Conference Proceedings / Presentations

(IC4) Taesoo Kim, Junghyun Seo, **Hyeonjae Jeon**, JinUng An, GyeungHo Choi, YongSeob Lim, "Lab to Real: CIELAB Loss for Realistic Night to Day Image Translation Model", 19th International Conference on Intelligent Autonomous Systems (IAS-19), Genoa, Italy, 2025.

(IC3) Junghyun Seo*, **Hyeonjae Jeon***, Joonyoung Choi, Kwangho Woo, Yongseob Lim, and Yongsik Jin, "BEV Image-based Lane Tracking Control System for Autonomous Lane Repainting Robot", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Abu Dhabi, UAE, 2024. (* equally contributed)

(IC2) Taesoo Kim, **Hyeonjae Jeon** and Yongseob Lim, "Challenges of YOLO-Series for Object Detection in Extremely Heavy Rain: CARLA Simulator-based Synthetic Evaluation Dataset", IEEE International Conference on Robotics and Automation (ICRA), LBR-Poster, London, UK, 2023.

(IC1) **Hyeonjae Jeon**, YaeOhn Kim, Minyoung Choi, Donggeon Park, Sungho Son, Jungki Lee, Gyeungho Choi, Yongseob Lim, "CARLA Simulator-based Evaluation Framework Development of Lane Detection Accuracy Performance under Sensor Blockage caused by Heavy Rain for Autonomous Vehicle", IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Kyoto, Japan, 2022.

Domestic Conference Presentations: Robot Perception and Control Systems

(DC7) Junghyun Seo, **Hyeonjae Jeon**, Yongsik Jin, & Yongseob Lim. (2023, November). Evaluation of forward/downward view camera-based lane detection and integrated autonomous driving control system. Institute of Embedded Engineering of Korea (IEMEK) Conference (Fall), Jeju, Korea. Outstanding Paper Presentation Award.

(DC6) Yaeohn Kim, **Hyeonjae Jeon**, Minyoung Choi, Jeseok Kim, Soon Kwon, Jinung An, Gyeungho Choi, & Yongseob Lim. (2022). Improved collision avoidance performance by using infra-surveillance sensor detecting objects in blind spots for autonomous mobile robot. The 36th ICROS Annual Conference, Geoje, Korea.

(DC5) Yaeohn Kim, **Hyeonjae Jeon.**, Minyoung Choi, Sungho Son, Jungki Lee, Jinung An, Gyeungho Choi, & Yongseob Lim. (2021, October 27-29). Improved lane changing control system with varying look-ahead distance in accordance with longitudinal vehicle velocity. The Conference of Korea Automotive Safety Association, Yeosu, Korea.

Domestic Conference Presentations: Adverse Weather and Autonomous Driving Safety

(DC4) Taesoo Kim, **Hyeonjae Jeon,** Junghyun Seo, & Yongseob Lim. (2023, May 25). Effects of adverse weather

conditions on object detection and time-to-collision estimation. 2023 Korea Society of Automotive Engineers (KSAE) Spring Conference, Pyeongchang, Korea, p. 943.

(DC3) Hyeonjae Jeon, Minyoung Choi, Donggeon Park, Sungho Son, Jungki Lee, Gyeungho Choi, & Yongseob Lim. (2022). Performance evaluation method of lane detection system under sensor blockage caused by adverse weather condition for autonomous vehicle. The Conference of Korea Automotive Safety Association, Yangyang, Korea.

Domestic Conference Presentations: Data Generation and Visual Representation

(DC2) Junghyun Seo, Sungjun Wang, **Hyeonjae Jeon**, Taesoo Kim, & Yongseob Lim. (2023, May 25). Adjusting the brightness of generated image for data augmentation in diverse night environment. 2023 Korea Society of Automotive Engineers (KSAE) Spring Conference, Pyeongchang, Korea, pp. 941-942.

(DC1) Sungjun Wang, Junghyun Seo, **Hyeonjae Jeon**, Sungjin Lim, Sanghyun Park, & Yongseob Lim. (2023, June 22). Epipolar line-based horizontal attention for domain adaptation of stereo matching algorithms. The 38th ICROS Annual Conference, pp. 321-322.

■ Technical Skills

- Robotics and simulation: ROS/ROS Noetic, CARLA, autonomous driving simulation, sensor integration, embedded computing board-based robot implementation.
- Perception and deep learning: lane detection, object detection, multi-task perception, YOLO-based detection, image-to-image translation, diffusion restoration, degradation-aware representation learning.
- Evaluation metrics: RMSE, lane-center deviation, steering angle error, PSNR, SSIM, LPIPS, FID, mAP, IoU, and task-oriented perception/driving metrics.
- Datasets and environments: BDD100K, Rain100H, LOL, GoPro, RESIDE-6k, custom BEV lane datasets, and CARLA-ROS driving scenarios.

■ Awards & Honors

(H1) Outstanding Paper Presentation Award, Paper Title: Evaluation of Forward/Downward View Camera-Based Lane Detection and Integrated Autonomous Driving Control System, Fall Conference in Institute of Embedded Engineering of Korea (IEMEK), November 2023.